

Project Title: Pulse Portal

Team members

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Project Overview

Objective

The objective of Pulse Portal is to develop an intelligent healthcare support system that enhances patient access to healthcare services and improves consultation management for doctors and professionals. By integrating structured appointment workflows with AI-assisted guidance, summarization, and administrative support, the system aims to facilitate informed decision-making, reduce communication barriers, and improve overall operational efficiency.

Target Audience

- **Patients** seeking a simple and organized platform to request healthcare consultations, manage appointments, and understand medical symptoms better.
- **Doctors and professionals** who require a structured system to manage consultations, review patient histories, and document visits.
- **Hospital administrative staff** responsible for coordinating appointment requests, scheduling consultations, and managing doctor availability.

Tech Stack

1. Backend

- **Laravel (PHP)**

2. Frontend

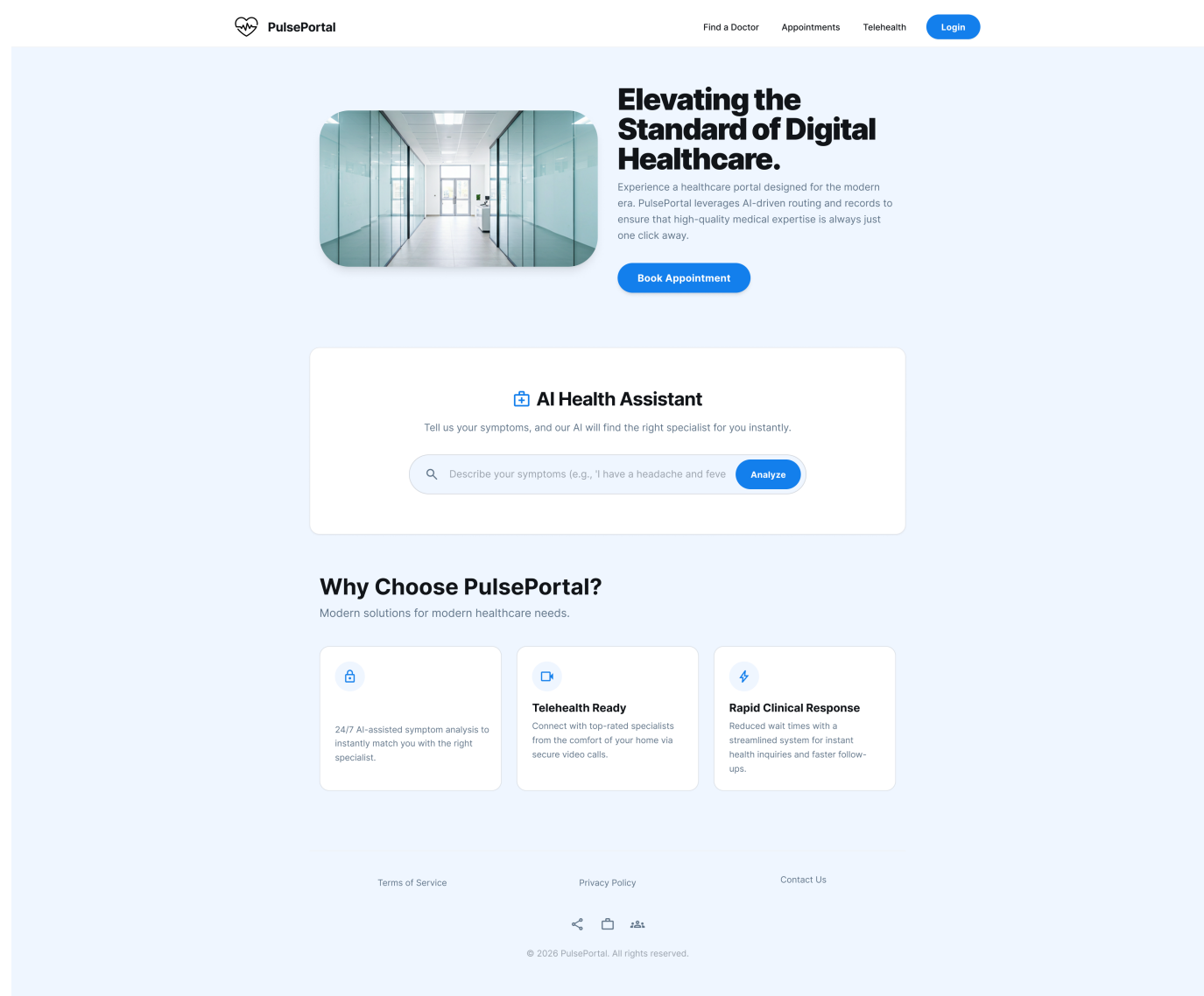
- **React**
- **TailwindCSS/ Bootstrap / MUI**

3. Rendering Method

- **Client-Side Rendering (CSR):** Adopted to provide a smooth, interactive experience for role-based users without the need for SEO.
4. Database
- **MySQL:** Will store patient profiles, doctor information, appointments, visit records, and AI-generated summaries.
5. AI Integration
- **OpenAI / Gemini API:** Will be used for AI-assisted features such as recommending specialists based on symptoms, generating visit summaries, and providing decision-support guidance.
6. Supporting Tools
- **JWT:** For secure authentication and API communication.

UI Mockups

Landing Page



Patient Dashboard

 PulsePortal

AppointmentsVisit HistoryPatient Dashboard

Patient Dashboard

Welcome back, Sarah.



Appointments

View all

UPCOMING

Oct 24, 10:00 AM



Dr. Emily Chen

Cardiology Dept.

Reschedule



PENDING

Nov 12, 02:30 PM



Dr. Michael Smith

General Practice

Details



Book Appointment



Visit History



Edit Profile



AI Health Support

Symptom checker & specialist finder

How are you feeling today? Describe your symptoms...

Find Relevant Specialist

Recent Visits

Your last medical consultations

...

Dr. Emily Chen

Dermatology • Sep 15, 2023

AI Summary

Dr. Marcus Thorne

General Practice • Aug 22, 2023

AI Summary

Dr. Olivia Reed

Ophthalmology • Jul 10, 2023

AI Summary

Appointment Booking

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PulsePortal

Appointments

Visit History

Patient Dashboard

How are you feeling today?

Our AI assistant will help you find the right specialist for your needs.

Q

Describe symptoms or search specialist name

AI Recommendation

Dr. Smith (Cardiology)

Based on your recent history and symptoms.

Select Doctor

Browse by Department

Select a category if you already know what you're looking for.

General Medicine

Cardiology

Pediatrics

Orthopedics

Neurology

Suggested Specialists

JD

John Doe

Cardiology

JD

John Doe

Cardiology

JD

John Doe

Cardiology

JD

John Doe

Cardiology

Appointment Details

SELECT DATE

October 2023

Su

Mo

Tu

We

Th

Fr

Sa

24

25

26

27

28

29

30

1

2

3

4

5

6

7

CONSULTATION MODE

In-Person

Online

AVAILABLE SLOTS

09:00 AM

10:30 AM

01:00 PM

02:30 PM

04:00 PM

Ready to confirm?

Doctor:

Dr. Jane Smith

Department:

Cardiology

Date & Time:

Oct 3, 2023 at 09:00 AM

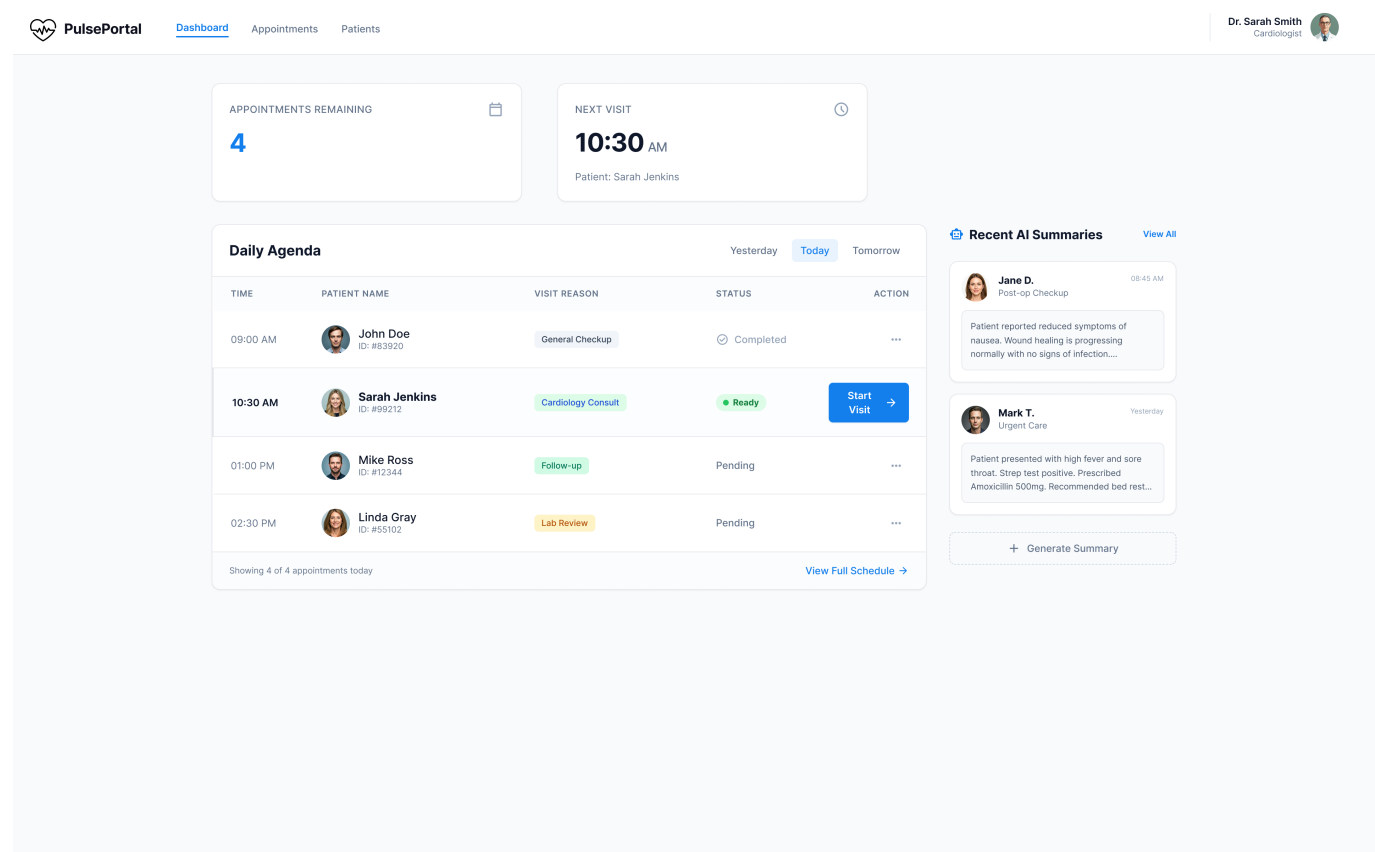
Type:

In-Person Consultation

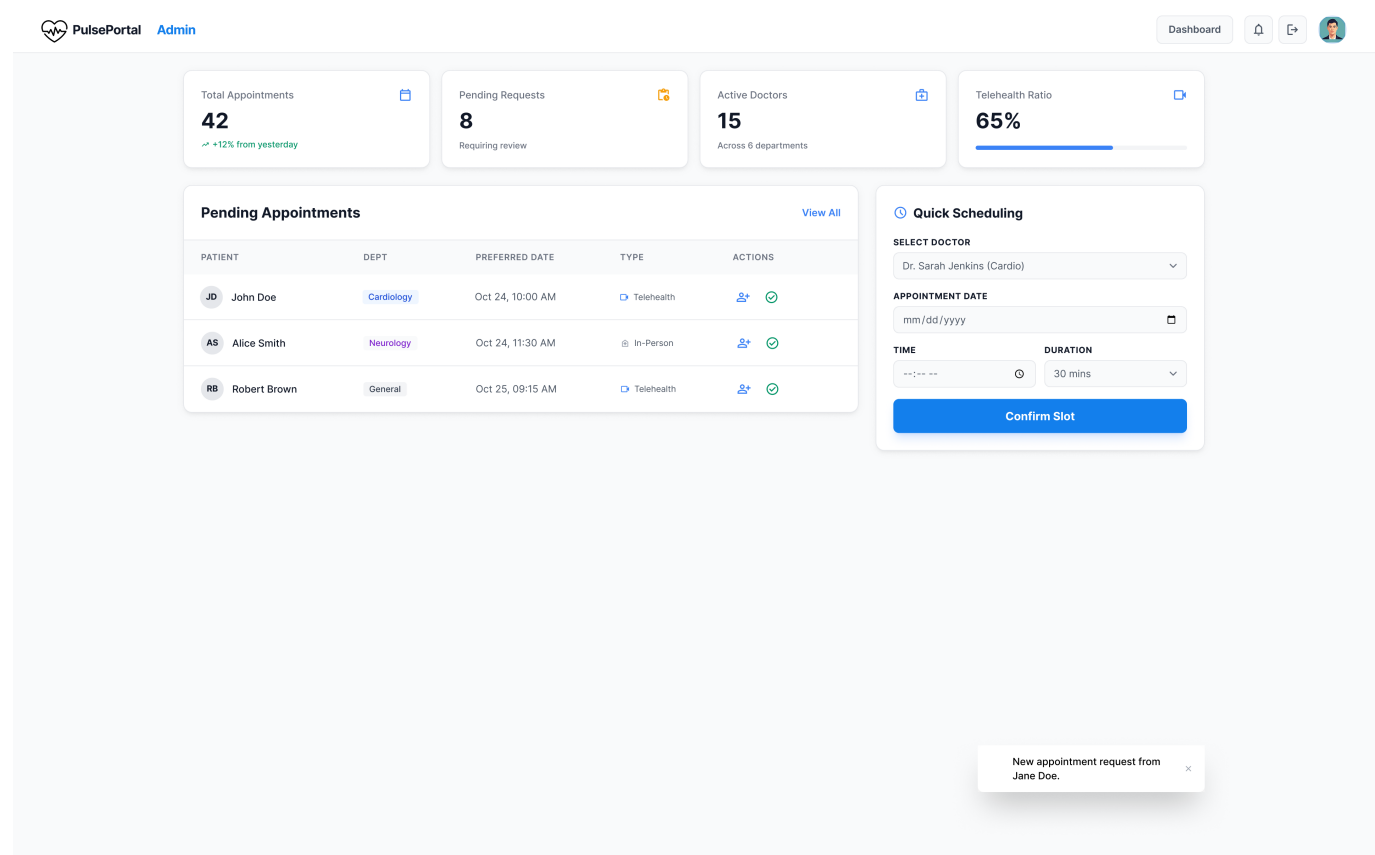
Confirm Booking

Doctor Dashboard

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Admin Dashboard



Figma Link: <https://www.figma.com/design/2R43AosEDGOxKwGeT3RqdJ/PulsePortal?m=auto&t=rdkGS48Rce4pU5TD-1>

Project Features

1. Main Features

- **Patient Management:** Patient registration, profile updates, appointment requests, and access to visit history.
- **Doctor Dashboard:** View assigned appointments, patient history access, and visit notes.
- **Appointment Management:** Structured appointment request, review, and support for both in-person and online consultations.
- **Role-Based Access Control:** Secure authentication and authorization for patients, doctors, and hospital staff based on assigned roles.

2. AI Integrated Features

- **Specialist Recommendation:** Suggest appropriate departments or specialists based on patient-reported symptoms.
- **Visit Summarization:** Generate simplified, patient-friendly summaries from doctor notes.
- **Past Visits Summarization:** Provide concise summaries of past visits to help doctors better understand their patients medical history.

| AI features are designed for understanding and decision-making and do not provide medical diagnoses or treatment recommendations.

3. CRUD Operations

- **Patients:** Create and update patient profiles and retrieve visit-related information.
- **Doctors:** Maintain doctor information including availability and specialization.
- **Appointments:** Create appointment requests, view schedules, update appointment status.
- **Visit Notes:** Record and retrieve visit notes and summaries added by doctors.

4. Key API Endpoints (approx.)

- **POST /auth/register** – Patient registration
- **POST /auth/login** – Login for all user roles
- **GET /patients/appointments** – View patient appointments
- **POST /patients/appointments** – Request a new appointment
- **GET /doctors/appointments** – View doctor scheduled appointments
- **POST /doctors/visit-notes** – Add visit notes
- **GET /patients/visit-history** – Fetch visit summaries

- **PUT /admin/appointments/{id}/schedule** – Admin schedules appointments
- **POST /admin/doctors** – Add doctor accounts
- **POST /patients/ai/specialist-recomm** – Suggest specialists based on symptoms
- **POST /doctors/ai/visit-summary** – Generate simplified summaries from past visit notes.

Milestones

Milestone 1: Core System Setup & Authentication

- Database design and initial schema (patients, doctors, admins, appointments)
- JWT-based authentication and role-based access control
- Patient registration and unified login system
- Design basic dashboards for patients, doctors and admins
- API integration between frontend and backend

Milestone 2: Appointment & Consultation Management

- Patient appointment request and management
- Doctor appointment viewing and status updates
- Admin appointment reviewing
- Visit history and consultation record storage

Milestone 3: AI Integration & Finalization

- Basic online consultation support
- AI-assisted specialist recommendation based on symptoms
- Historical visit summarization for patients
- UI enhancements and improved user experience
- Testing, bug fixing, and final deployment preparation